

Question 1: Posterior Probability and Bayes Classification

Suppose there are two classes, ω_1 and ω_2 , with the following parameters:

- Prior probabilities:
 $P(\omega_1) = 0.6$,
 $P(\omega_2) = 0.4$
- Class-conditional probability densities (likelihoods) for a given feature value $x = 4$:
 $p(x = 4 | \omega_1) = 0.2$
 $p(x = 4 | \omega_2) = 0.5$

Tasks:

- Compute the **posterior probabilities** $P(\omega_1 | x = 4)$ and $P(\omega_2 | x = 4)$ using Bayes' theorem.
- Based on the posterior probabilities, assign the class label to $x = 4$ using the **Bayes decision rule**.
- What is the **probability of classification error** at $x = 4$?

① $P(\omega_1) = 0.6$ $P(x=4|\omega_1) = 0.2$
 $P(\omega_2) = 0.4$ $P(x=4|\omega_2) = 0.5$

② Posterior Probabilities
By Bayes' theorem, $P(\omega_i|x) = \frac{P(\omega_i) \cdot P(x|\omega_i)}{P(x)}$

$$P(x=4) = P(\omega_1) \cdot P(x|\omega_1) + P(\omega_2) \cdot P(x|\omega_2)$$
$$= 0.6 \times 0.2 + 0.4 \times 0.5 = 0.12 + 0.2 = 0.32$$
$$\therefore P(\omega_1|x=4) = \frac{0.6 \times 0.2}{0.32} = 0.375$$
$$P(\omega_2|x=4) = \frac{0.4 \times 0.5}{0.32} = 0.625$$

③ As $P(\omega_2|x=4) > P(\omega_1|x=4) = 0.625 > 0.375$.
 $\therefore$ We choose the class ω_2 with higher posterior.

④ Probability of error (P_E) = $\min(P(\omega_1|x), P(\omega_2|x))$
 $= 0.375$.

⑤ $g_1(3) = \ln 0.4 - \frac{(3-2)^2}{2} = -1.4163$
 $g_2(3) = \ln 0.6 - \frac{(3-5)^2}{2} = -2.5108$
As $g_1(3) > g_2(3) \rightarrow$ we assign class ω_1 .

Question 2: Bayes' Decision Rule for Continuous Case (Gaussian)

Let $x \in \mathbb{R}$ be a one-dimensional continuous feature. Consider two classes ω_1 and ω_2 with the following distributions:

- $p(x|\omega_1) \sim \mathcal{N}(2, 1)$
- $p(x|\omega_2) \sim \mathcal{N}(5, 1)$
- Prior probabilities: $P(\omega_1) = 0.4$, $P(\omega_2) = 0.6$

Tasks:

- Derive the **discriminant functions** $g_1(x)$ and $g_2(x)$.
- Find the **decision boundary** (value of x) at which $g_1(x) = g_2(x)$.
- For $x = 3$, determine which class is assigned by the Bayes classifier.

② $\omega_1: \mathcal{N}(2, 1)$ means $\mu_1 = 2, \sigma^2 = 1$
 $\omega_2: \mathcal{N}(5, 1)$ means $\mu_2 = 5, \sigma^2 = 1$
 $P(\omega_1) = 0.4$
 $P(\omega_2) = 0.6$

③ Discriminant Fun.

In normal distribution, $p(x|\omega_i) = \frac{1}{\sqrt{2\sigma^2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$

By Bayes' theorem, $P(\omega_i|x) \propto P(\omega_i) \cdot P(x|\omega_i)$

Now let the discriminant function be,

$$g_i(x) = \ln(P(\omega_i)) + \ln(p(x|\omega_i))$$
$$\therefore = \ln(P(\omega_i)) + \ln\left(\frac{1}{\sqrt{2\pi\sigma^2}} \cdot e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}\right)$$
$$= \ln(P(\omega_i)) - \frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2 + \text{constant}$$
$$g_i(x) = \ln(P(\omega_i)) - \frac{(x-\mu)^2}{2\sigma^2}$$
$$\therefore \left. \begin{aligned} g_1(x) &= \ln(0.4) - \frac{(x-2)^2}{2} \\ g_2(x) &= \ln(0.6) - \frac{(x-5)^2}{2} \end{aligned} \right\} \text{ Here } \sigma = 1$$

$$\begin{aligned}
 \textcircled{b} \quad g_1(x) &= g_2(x) \Rightarrow \ln(0.4) - \frac{(x-2)^2}{2} = \ln(0.6) - \frac{(x-5)^2}{2} \\
 &\Rightarrow \frac{(x-5)^2 - (x-2)^2}{2} = \ln(0.6 \div 0.4) \\
 &\Rightarrow x^2 - 10x + 25 - x^2 + 4x - 4 = 2 \ln(1.5) \\
 &\Rightarrow -6x + 21 = 2 \ln(1.5) \Rightarrow x = \frac{(2 \ln(1.5) - 21)}{6} \\
 &\Rightarrow x = 3.3648
 \end{aligned}$$

$$\begin{aligned}
 \textcircled{c} \quad g_1(3) &= \ln 0.4 - \frac{(3-2)^2}{2} = -1.4163 \\
 g_2(3) &= \ln 0.6 - \frac{(3-5)^2}{2} = -2.5108 \\
 \text{As } g_1(3) &> g_2(3) \longrightarrow \text{we assign class } \omega_1
 \end{aligned}$$

Question 3: Posterior Using Multivariate Gaussian

You are given a two-class classification problem with:

- $\omega_1 : \mu_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \Sigma_1 = I$
- $\omega_2 : \mu_2 = \begin{bmatrix} 2 \\ 2 \end{bmatrix}, \Sigma_2 = I$
- Priors: $P(\omega_1) = 0.5, P(\omega_2) = 0.5$

Let the feature vector be $x = \begin{bmatrix} 1.5 \\ 1 \end{bmatrix}$

Tasks:

- Write the general form of the **discriminant function** for multivariate Gaussians with equal covariance.
- Compute $g_1(x)$ and $g_2(x)$.
- Classify x based on the discriminant functions.

$$\textcircled{3} \quad \mu_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \Sigma_1 = I \quad P(\omega_1) = 0.5 \quad x = \begin{bmatrix} 1.5 \\ 1 \end{bmatrix} \\ \mu_2 = \begin{bmatrix} 2 \\ 2 \end{bmatrix}, \Sigma_2 = I \quad P(\omega_2) = 0.5$$

② For Multivariate Gaussians with equal covariances,

$$p(x|\omega_i) = \frac{1}{\sqrt{(2\pi)^k |\Sigma|}} e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1} (x-\mu)}$$

$$\boxed{g_i(x) = \ln(p(\omega_i)) + \ln(p(x|\omega_i))} \quad \text{Discriminant func.}$$

$$\ln(p(x|\omega_i)) = \ln\left(\frac{1}{\sqrt{(2\pi)^k}}\right) - \frac{1}{2}(x-\mu)^T (x-\mu)$$

$$= -\frac{1}{2}(x-\mu)^T (x-\mu) + \text{const.} \quad \left[\Sigma = I, \text{ so removed} \right]$$

$$\boxed{g_i(x) = \ln(p(\omega_i)) - \frac{1}{2}(x-\mu_i)^T (x-\mu_i)}$$

$$\textcircled{b} \quad (x-\mu_1) = \begin{bmatrix} 1.5 \\ 1 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0.5 \\ 0 \end{bmatrix}$$

$$(x-\mu_1)^T (x-\mu_1) = \begin{bmatrix} 0.5 & 0 \end{bmatrix} \begin{bmatrix} 0.5 \\ 0 \end{bmatrix} = 0.25$$

$$(x-\mu_2) = \begin{bmatrix} 1.5 \\ 1 \end{bmatrix} - \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} -0.5 \\ -1 \end{bmatrix}$$

$$(x-\mu_2)^T (x-\mu_2) = \begin{bmatrix} -0.5 & -1 \end{bmatrix} \begin{bmatrix} -0.5 \\ -1 \end{bmatrix} = 0.25 + 1 = 1.25$$

$$\therefore g_1(x) = \ln(0.5) - \frac{1}{2}(0.25) = -0.8181$$

$$g_2(x) = \ln(0.5) - \frac{1}{2}(1.25) = -1.3181$$

③ As $g_1(x) > g_2(x)$, $\therefore x$ belongs to ω_1

Q6. Show that Bayes Classifier minimizes the probability of error

④ Show that Bayes classifier minimizes probability of error

Let

- $w_1, w_2, \dots, w_n$ be the possible classes
- x be the observed feature vector
- $P(w_i/x)$ be the posterior probability of class w_i given by observation x

So, if Bayes classifier observes x to be the class with highest posterior probability then,

$$w^* = \underset{w_i}{\operatorname{argmax}} P(w_i/x) \quad \text{--- (i)}$$

Now, we need to show that w^* that maximizes $P(w_i/x)$ also minimizes probability of error.

Error Definition for x

→ If x is assigned to class w_i , the probability that it is correct (without error) $\phi = P(w_i/x)$

→ And the probability of error (assigned to wrong class w_j)

$$P_e = 1 - P(w_i/x)$$

Maximization of $P(w_i/x)$

Now, Bayes classifier looks at all possible classes

$w_1, w_2, \dots, w_n$, & picks the highest posterior probability which is (i)

$\therefore P(w_i/x)$ is maximized.

$\therefore P_e = 1 - P(w_i/x)$ is minimized

Thus error got minimized.

Q5. Text Classification using Naïve Bayes

You are given the following dataset for spam detection:

Email ID	Contains "win"	Contains "money"	Label
1	Yes	Yes	Spam
2	Yes	No	Spam
3	No	Yes	Not Spam
4	No	No	Not Spam

Using Naïve Bayes classifier, classify a new email that contains both "win" and "money".

Assume Laplace smoothing and calculate:

- a) Prior probabilities
- b) Conditional probabilities
- c) Posterior for both classes
- d) Final classification

5.a) Prior Probabilities

$$P(\text{Spam}) = 2/4 = 0.5$$
$$P(\text{not Spam}) = 2/4 = 0.5$$

5.b) Conditional Probabilities (using Laplace smoothing)

SPAM

$$P(\text{win} = \text{Yes} | \text{SPAM}) = \frac{\text{count of Yes} + 1}{\text{total sample} + K} \quad (K = \text{Yes or No} = 2)$$
$$= \frac{2 + 1}{2 + 2} = 0.75$$
$$P(\text{money} = \text{Yes} | \text{SPAM}) = \frac{1 + 1}{2 + 2} = 0.5$$

NOT SPAM

$$P(\text{win} = \text{Yes} | \text{NOT SPAM}) = \frac{0 + 1}{4} = 0.25$$
$$P(\text{Money} = \text{Yes} | \text{not Spam}) = \frac{1 + 1}{4} = 0.5$$

5.c) Posterior probability

$$P(\text{Spam} | x) \propto P(\text{Spam}) \cdot P(\text{win} = \text{Yes} | \text{Spam}) \cdot P(\text{money} = \text{Yes} | \text{Spam})$$
$$\therefore P(\text{Spam} | x) \propto (0.5 \times 0.75 \times 0.5) = \frac{3}{16}$$
$$P(\text{not spam} | x) \propto (0.5 \times 0.25 \times 0.5) = \frac{1}{16}$$

5.d) Conclusion

$$P(\text{Spam} | x) > P(\text{not Spam} | x)$$

$\therefore x$ is a Spam email.

Q6. Gaussian (Normal) Naïve Bayes for Continuous Features

You are given the following data for a **binary classification problem** (classes ω_1 and ω_2) with **two continuous features**, x_1 and x_2 . Assume features follow **Gaussian distribution** and are **conditionally independent**.

Class (ω)	x_1 Mean	x_1 Variance	x_2 Mean	x_2 Variance
ω_1	2	1	3	4
ω_2	4	1	1	4

The prior probabilities are:

- $P(\omega_1) = 0.6$
- $P(\omega_2) = 0.4$

You are given a test point: $x = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$

Classify the point using the **Gaussian Naïve Bayes classifier**.

⑥ By conditional probability,

$$P(\omega_i|x) = \frac{P(\omega_i) P(x|\omega_i)}{P(x)} \propto P(\omega_i) \cdot P(x|\omega_i) \quad \text{--- (i)}$$

$$P(x|\omega_i) = \prod_{j=1}^n P(x_j|\omega_i) \quad \text{--- (ii) From Naive Bayes classifier}$$

$P(x_j|\omega_i)$ for Normal/Gaussian Distribution

$$P(x_j|\omega_i) = \frac{1}{\sqrt{2\pi\sigma_j^2}} e^{-\frac{(x_j - \mu_{ij})^2}{2\sigma_j^2}} \quad \text{--- (iii)}$$

$\therefore$ From (i), (ii), (iii)

$$P(\omega_i|x) \propto P(\omega_i) \prod_{j=1}^n \frac{1}{\sqrt{2\pi\sigma_j^2}} e^{-\frac{(x_j - \mu_{ij})^2}{2\sigma_j^2}}$$

Now, Class w_1 $P(w_1) = 0.6$ $x = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$

$\mu_{11} = 2, \sigma_{11}^2 = 1$

$$P(x_1|w_1) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{3-2}{1} \right)^2}$$

$$= \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}} = 0.2419$$

$$P(x_2|w_1) = \frac{1}{\sqrt{2\pi \cdot 4}} e^{-\frac{1}{2} \left(\frac{2-3}{2} \right)^2}$$

$$= \frac{1}{\sqrt{8\pi}} e^{-\frac{1}{8}} = 0.1750$$

Now Class w_2

$$P(x_1|w_2) = \frac{1}{\sqrt{2\pi \cdot 1}} e^{-\frac{(3-4)^2}{2 \cdot 1}} = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}} = 0.2419$$

$$P(x_2|w_2) = \frac{1}{\sqrt{2\pi \cdot 4}} e^{-\frac{(2-1)^2}{2 \cdot 4}} = \frac{1}{\sqrt{8\pi}} e^{-\frac{1}{8}} = 0.1760$$

Now Likelihood

$$w_1 / P(x|w_1) = 0.2419 \times 0.1760 = 0.04257$$

$$P(w_1|x) \propto P(w_1) P(x|w_1) = 0.6 \times 0.04257 = 0.02554$$

$$w_2 / P(x|w_2) = 0.2419 \times 0.1760 = 0.04257$$

$$P(w_2|x) \propto P(w_2) P(x|w_2) = 0.4 \times 0.04257 = 0.01703$$

$$\therefore \text{As } P(w_1|x) > P(w_2|x)$$

$\therefore x$ belongs to w_1 class.

Q7. Numerical - Gaussian Naïve Bayes Classifier

A bank is building a model to predict whether a customer will **default** on a loan. You are given the following data:

Class (Default = Yes/No)	Income Mean (in lakhs)	Income Variance	Age Mean	Age Variance
Yes	3	1	40	25
No	6	4	30	36

The prior probabilities are:

- $P(\text{Default} = \text{Yes}) = 0.3$
- $P(\text{Default} = \text{No}) = 0.7$

A new customer has:

- Income = 4.5 lakhs
- Age = 35

Using **Gaussian Naïve Bayes**, determine whether the customer will default.

⑦ $P(x_j | w_i) = \frac{1}{\sqrt{2\pi\sigma^2}} \cdot e^{-\frac{(x_j - \mu_j)^2}{2\sigma^2}}$ } From gaussian (normal) distribution.

For class Yes

Income $P(4.5 | \text{Yes}) = \frac{1}{\sqrt{2\pi \cdot 1}} \cdot e^{-\frac{(4.5 - 3)^2}{2 \cdot 1}} = \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{2.25}{2}} = 0.12949$

Age $P(35 | \text{Yes}) = \frac{1}{\sqrt{2\pi \cdot 25}} \cdot e^{-\frac{(35 - 40)^2}{2 \cdot 25}} = \frac{1}{\sqrt{50\pi}} \cdot e^{-\frac{1}{2}} = 0.04838$

$\therefore P(x | \text{Yes}) = 0.12949 \times 0.04838 = 0.0062647$

$P(\text{Yes} | x) \propto P(x | \text{Yes}) \times P(\text{Yes}) = 0.0062647 \times 0.3 = \boxed{1.8794 \times 10^{-3}}$

For class NO

Income $P(4.5 | \text{No}) = \frac{1}{\sqrt{2\pi \cdot 4}} \cdot e^{-\frac{(4.5 - 6)^2}{2 \cdot 4}} = \frac{1}{\sqrt{8\pi}} \cdot e^{-0.28125} = 0.15054$

Age $P(35 | \text{No}) = \frac{1}{\sqrt{2\pi \cdot 36}} \cdot e^{-\frac{(35 - 30)^2}{2 \cdot 36}} = \frac{1}{\sqrt{72\pi}} \cdot e^{-0.34722} = 0.046976$

$P(x | \text{No}) = 0.15054 \times 0.046976 = 7.071767 \times 10^{-3}$

$P(\text{No} | x) \propto P(x | \text{No}) \cdot P(\text{No}) = 7.071767 \times 10^{-3} \times 0.7 = \boxed{4.950236 \times 10^{-3}}$

$P(\text{No} | x) > P(\text{Yes} | \text{No}) \therefore$ No Default for new customer.

Q8. [Application + Numerical]

You are building a spam filter using Naïve Bayes. The vocabulary consists of the words:
{free, win, now, offer, hello}

From training data, you calculate the following (word counts with Laplace smoothing):

Word	Spam Count	Not Spam Count
free	4	1
win	3	0
now	2	2
offer	5	1
hello	0	3

Class priors:

- $P(\text{Spam}) = 0.5$
- $P(\text{Not Spam}) = 0.5$

Total words (Spam) = 20, Not Spam = 15

Classify the email: "free offer now" using Multinomial Naïve Bayes.

Show each step clearly and compute posterior probabilities.

⑧ Vocabulary size = 5
 $V = \{ \text{free, win, now, offer, hello} \}$

$$P(w|\text{spam}) = \frac{\text{Count}(w) + 1}{\text{Total spam} + V} = \frac{\text{Count}(w) + 1}{20 + 5}$$

word	count	$P(w \text{spam})$
free	4	$5/25 = 1/5$
offer	5	$6/25$
now	2	$3/25$

$$P(w|\text{not spam}) = \frac{\text{Count}(w) + 1}{\text{Total not spam} + V} = \frac{\text{Count}(w) + 1}{15 + 5}$$

word	count	$P(w \text{not spam})$
free	1	$2/20 = 1/10$
offer	1	$2/20 = 1/10$
now	2	$3/20$

$$P(\text{Spam}|\text{email}) = 0.5 \times \frac{1}{5} \times \frac{6}{25} \times \frac{3}{25} = 0.288 \times 10^{-4}$$
$$P(\text{not spam}|\text{email}) = 0.5 \times \frac{1}{10} \times \frac{1}{10} \times \frac{3}{20} = 7.5 \times 10^{-4}$$

$\therefore$ As $P(\text{spam}|\text{email}) > P(\text{not spam}|\text{email})$
email is a spam.

Q9. [Derivation + Explanation]

Prove that when the covariance matrices $\Sigma_1 = \Sigma_2 = \Sigma$, the decision boundary becomes linear.

Start from the general discriminant function and show how the quadratic terms cancel, leading to a linear expression of the form:

$$g(x) = w^T x + w_0$$

Also, express w and w_0 in terms of $\mu_i, \Sigma, P(\omega_i)$.

Q The covariance matrices are equal, $\therefore \Sigma_1 = \Sigma_2 = \Sigma$

Considering general form of discriminant func,

$$g_i(x) = \ln(p(\omega_i)) + \ln(p(x|\omega_i))$$

From Multivariate gaussian distribution,

$$p(x|\omega_i) = \frac{1}{\sqrt{(2\pi)^k |\Sigma|}} e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1} (x-\mu)}$$

$$\therefore g_i(x) = \ln(p(\omega_i)) + \ln\left(\frac{1}{\sqrt{(2\pi)^k |\Sigma|}}\right) - \frac{1}{2}(x-\mu)^T \Sigma^{-1} (x-\mu)$$

$$g_i(x) = \ln(p(\omega_i)) - \frac{1}{2}(x-\mu)^T (\Sigma^{-1}) (x-\mu) \quad \text{constant eliminated.}$$

Let's consider the decision rule, $g_1(x) > g_2(x)$

$$\begin{aligned} \therefore g_1(x) - g_2(x) &= \ln(p(\omega_1)) - \frac{1}{2}(x-\mu_1)^T \Sigma^{-1} (x-\mu_1) - \\ &\quad \ln(p(\omega_2)) + \frac{1}{2}(x-\mu_2)^T \Sigma^{-1} (x-\mu_2) \\ &= \ln\left(\frac{p(\omega_1)}{p(\omega_2)}\right) - \frac{1}{2} \left[(x-\mu_1)^T \Sigma^{-1} (x-\mu_1) - (x-\mu_2)^T \Sigma^{-1} (x-\mu_2) \right] \end{aligned}$$

Now, we expand the inner parts.

$$\begin{aligned} (x-\mu_1)^T \Sigma^{-1} (x-\mu_1) &= x^T \Sigma^{-1} x - x^T \Sigma^{-1} \mu_1 - \mu_1^T \Sigma^{-1} x + \mu_1^T \Sigma^{-1} \mu_1 \\ &= x^T \Sigma^{-1} x - 2\mu_1^T \Sigma^{-1} x + \mu_1^T \Sigma^{-1} \mu_1 \end{aligned}$$

$$(x-\mu_2)^T \Sigma^{-1} (x-\mu_2) = x^T \Sigma^{-1} x - 2\mu_2^T \Sigma^{-1} x + \mu_2^T \Sigma^{-1} \mu_2$$

$$\begin{aligned} \therefore g_1(x) - g_2(x) &= \ln\left(\frac{p(\omega_1)}{p(\omega_2)}\right) - \frac{1}{2} \left[-2\mu_1^T \Sigma^{-1} x + \mu_1^T \Sigma^{-1} \mu_1 \right. \\ &\quad \left. + 2\mu_2^T \Sigma^{-1} x - \mu_2^T \Sigma^{-1} \mu_2 \right] \\ &= \ln\left(\frac{p(\omega_1)}{p(\omega_2)}\right) - \frac{1}{2} \left[2(\mu_2^T - \mu_1^T) \Sigma^{-1} x + (\mu_1^T \Sigma^{-1} \mu_1 - \mu_2^T \Sigma^{-1} \mu_2) \right] \end{aligned}$$

This is of the form, $g(x) = w^T x + w_0$ — Linear func.

$$w_0 = \ln\left(\frac{p(\omega_1)}{p(\omega_2)}\right) - \frac{1}{2}(\mu_1^T \Sigma^{-1} \mu_1 - \mu_2^T \Sigma^{-1} \mu_2) \quad w^T = (\mu_1^T - \mu_2^T) \Sigma^{-1}$$

Q10. [Numerical + Quadratic Discriminant Function]

Consider two classes with the following:

- $\mu_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $\Sigma_1 = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$
- $\mu_2 = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$, $\Sigma_2 = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$
- $P(\omega_1) = 0.5$, $P(\omega_2) = 0.5$

(a) Derive the quadratic discriminant functions for both classes.

(b) Compute the discriminant values for $x = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$, and classify the point.

⑩ $\mu_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ $\Sigma_1 = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$ $P(\omega_1) = 0.5$
 $\mu_2 = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$ $\Sigma_2 = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$ $P(\omega_2) = 0.5$

For Multivariate Gaussian Distribution,

$$p(x|\omega_i) = \frac{1}{\sqrt{(2\pi)^k |\Sigma|}} e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)}$$

$$g_i(x) = \ln(p(\omega_i)) + \ln\left(\frac{1}{\sqrt{(2\pi)^k |\Sigma|}}\right) - \frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)$$

$$\ln\left(\frac{1}{\sqrt{(2\pi)^k |\Sigma|}}\right) = \ln\left((2\pi)^{-k/2} (|\Sigma|)^{-1/2}\right)$$

$$= -\frac{k}{2} \ln(2\pi) - \frac{1}{2} \ln(|\Sigma|)$$

$$= -\frac{1}{2} \ln(|\Sigma|) \quad \text{--- const ignored}$$

$$\therefore g_i(x) = \ln(p(\omega_i)) - \frac{1}{2} \ln(|\Sigma|) - \frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)$$

$$|\Sigma_1| = 2 \quad |\Sigma_2| = 5$$

$$\therefore \Sigma_1^{-1} = \frac{1}{2} \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0.5 \end{bmatrix}$$

$$\Sigma_2^{-1} = \frac{1}{5} \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} = \frac{1}{5} \begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix}$$

$$\therefore (x-\mu_1)^T = \begin{bmatrix} 3 \\ 2 \end{bmatrix} - \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

$$(x-\mu_2)^T = \begin{bmatrix} 3 \\ 2 \end{bmatrix} - \begin{bmatrix} 4 \\ 3 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$$

$$\therefore (x - \mu_1)^T \Sigma^{-1} (x - \mu_1) = [2 \ 0] \begin{bmatrix} 1 & 0 \\ 0 & 0.5 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \end{bmatrix} \\ = [2 \ 0] \begin{bmatrix} 2 \\ 0 \end{bmatrix} = 4$$

$$(x - \mu_2)^T \Sigma^{-1} (x - \mu_2) = [-1 \ -1] \frac{1}{5} \begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} -1 \\ -1 \end{bmatrix} \\ = \frac{1}{5} [-1 \ -1] \begin{bmatrix} -2 \\ -1 \end{bmatrix} = \frac{3}{5}$$

$$g_1(x) = \ln(0.3) - \frac{1}{2} \ln(2) - \frac{1}{2}(4) - \cancel{0.69315} - \\ = -0.69315 - 0.34657 - 2 = -3.03972$$

$$g_2(x) = \ln(0.3) - \frac{1}{2} \ln(3) - \frac{1}{2} \left(\frac{3}{5}\right) \\ = -0.69315 - 0.804719 - 0.3 \\ = -1.7979$$

As $g_1(x) < g_2(x)$, x is assigned to w_1

Question 11

A dataset consists of single discrete features with 3 possible values: A, B, C. The class labels are w_1 & w_2 . Given following probabilities:

x	$P(x w_1)$	$P(x w_2)$	$P(w_1) = 0.4$ $P(w_2) = 0.6$
A	0.3	0.1	
B	0.5	0.6	
C	0.2	0.3	

- Find
- $P(w_1|x)$ & $P(w_2|x)$ for all $x \in \{A, B, C\}$
 - Construct optimal decision rub.
 - Compute probability ^{of error} of the classifier.

Ans)

a) $P(w_1|x) = \frac{P(x|w_1) P(w_1)}{P(x)}$

$$P(x) = P(x|w_1)P(w_1) + P(x|w_2)P(w_2)$$

$$P(A) = 0.3 \times 0.4 + 0.1 \times 0.6 = 0.18$$

$$P(B) = 0.5 \times 0.4 + 0.6 \times 0.6 = 0.56$$

$$P(C) = 0.2 \times 0.4 + 0.3 \times 0.6 = 0.26$$

$$P(w_1|A) = \frac{0.3 \times 0.4}{0.18} = 0.6667$$

$$P(w_2|A) = \frac{0.1 \times 0.6}{0.18} = 0.3333$$

$$P(w_1|B) = \frac{0.5 \times 0.4}{0.56} = 0.3571$$

$$P(w_2|B) = \frac{0.6 \times 0.6}{0.56} = 0.6429$$

$$P(w_1|C) = \frac{0.2 \times 0.4}{0.26} = 0.3077$$

$$P(w_2|C) = \frac{0.3 \times 0.6}{0.26} = 0.6923$$

⑤

x		$x \in \text{class}$
A	$P(x \omega_1) > P(x \omega_2)$	ω_1
B	$<$	ω_2
C	$<$	ω_2

⑥ Probability of error (P_e)

$$P_e = 1 - P(\omega_i|x) \quad \left\{ P(\omega_i|x) \text{ is max. posterior prob.} \right\}$$

$\therefore$

x	error
A	$1 - 0.6667 = 0.3333$
B	$1 - 0.6429 = 0.3571$
C	$1 - 0.6923 = 0.3077$